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1.mapping color

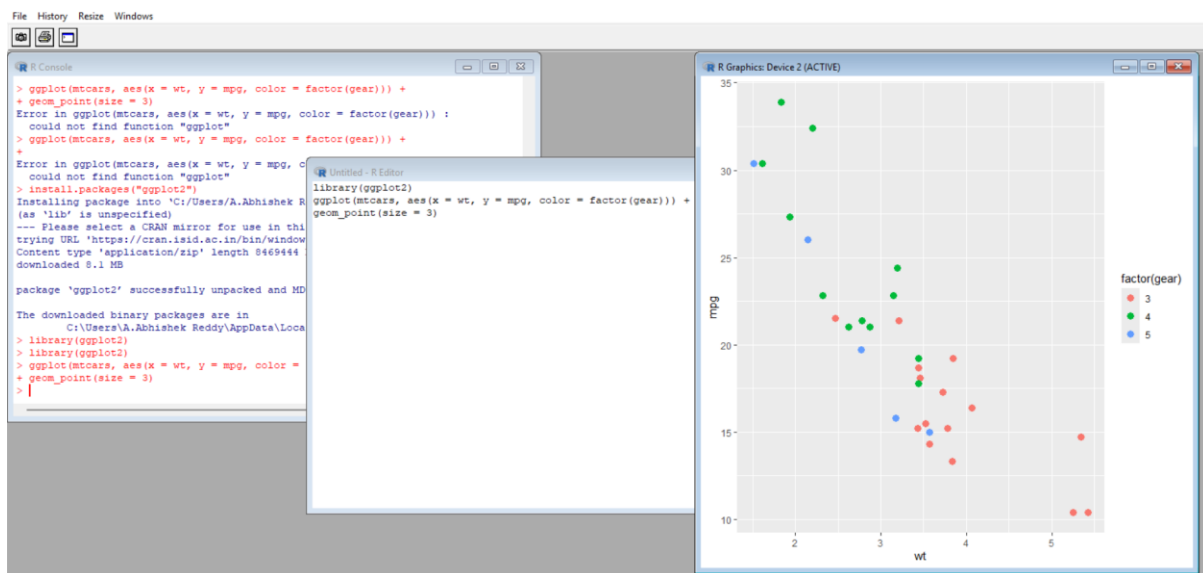
A. CODE

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, color = factor(gear))) +
```

```
geom_point(size = 3)
```

OUTPUT:



2.Continuous example:

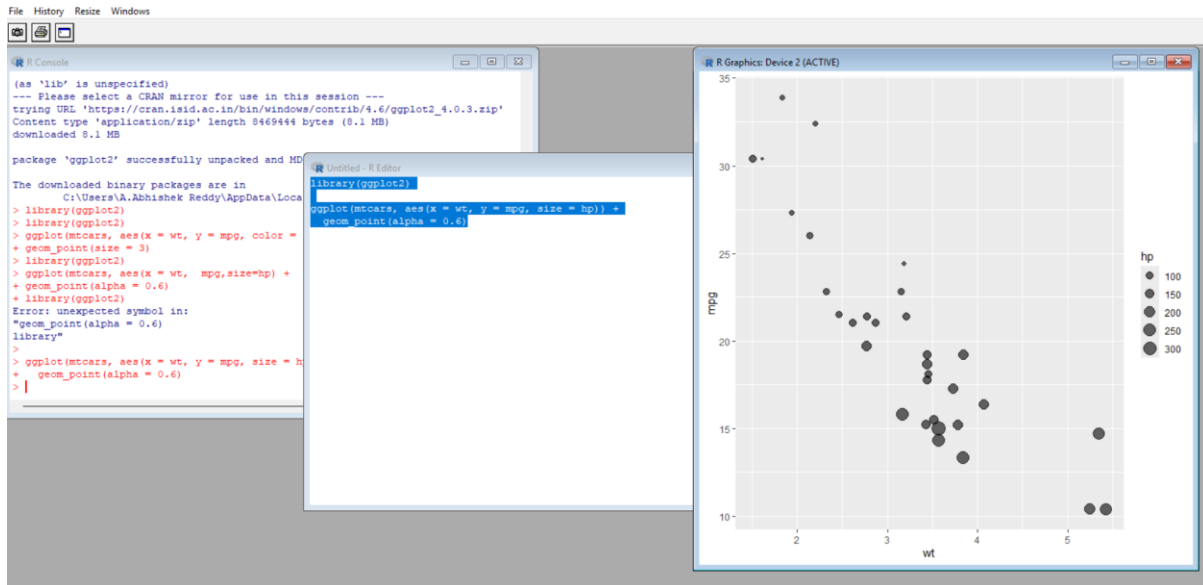
Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, size = hp)) +
```

```
geom_point(alpha = 0.6)
```

OUTPUT:



3.modify color size

Code:

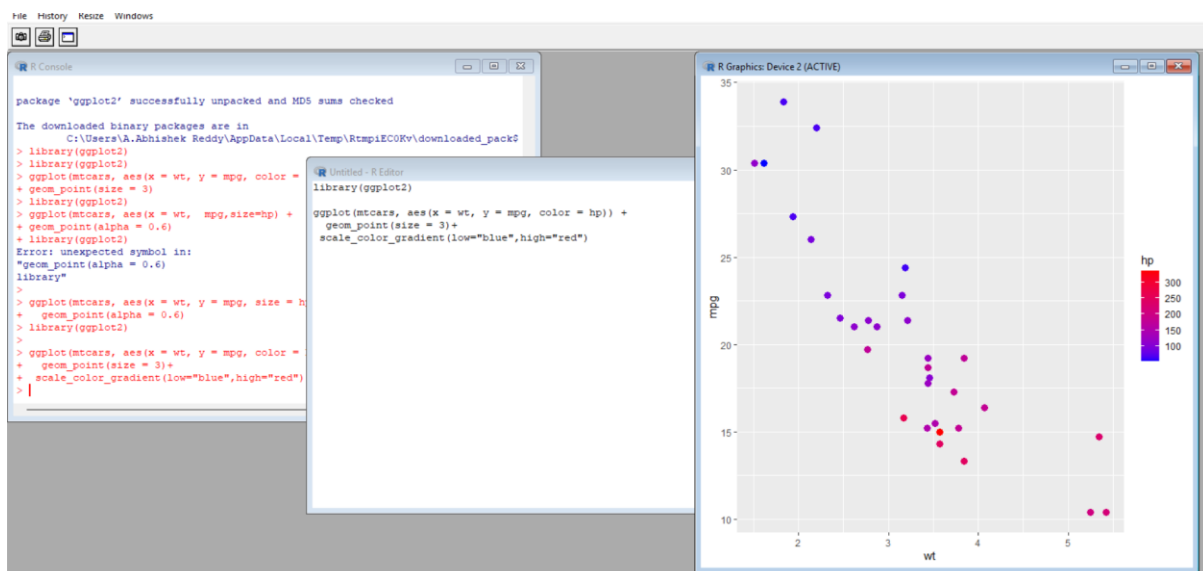
```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, color = hp)) +
```

```
  geom_point(size = 3)+
```

```
  scale_color_gradient(low="blue",high="red")
```

OUTPUT:



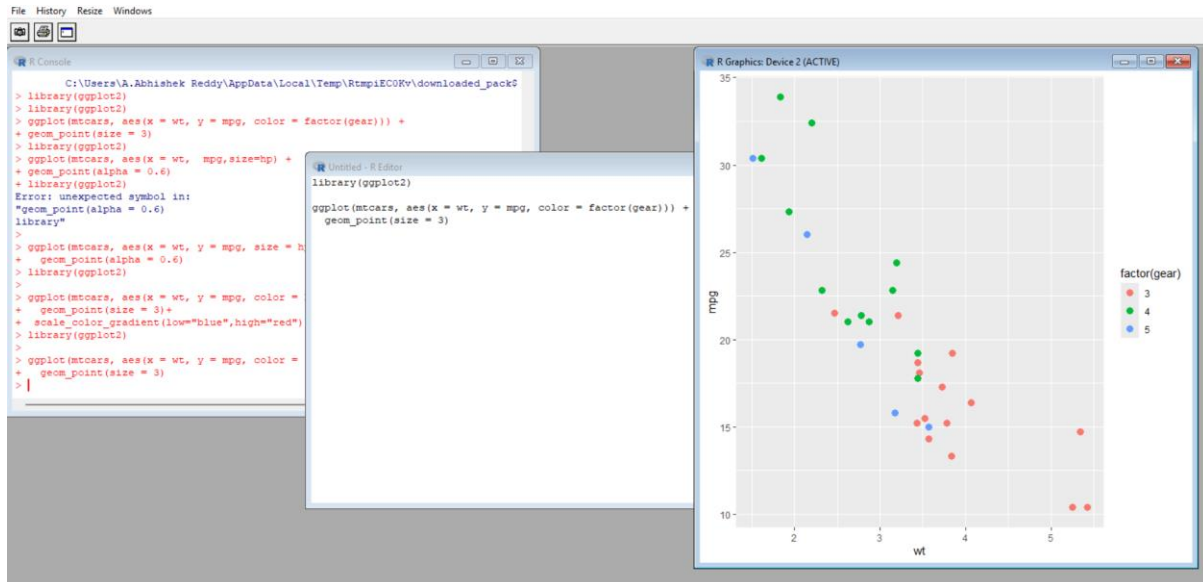
4.categorical example:

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, color = factor(gear))) +  
  geom_point(size = 3)
```

OUTPUT:



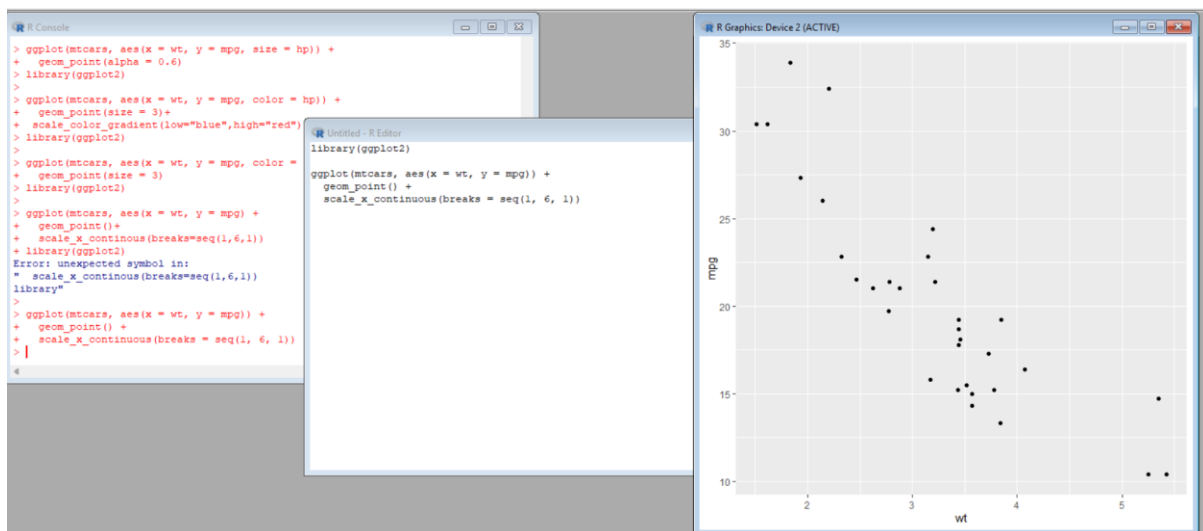
5.custom breaks

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  scale_x_continuous(breaks = seq(1, 6, 1))
```

Output:



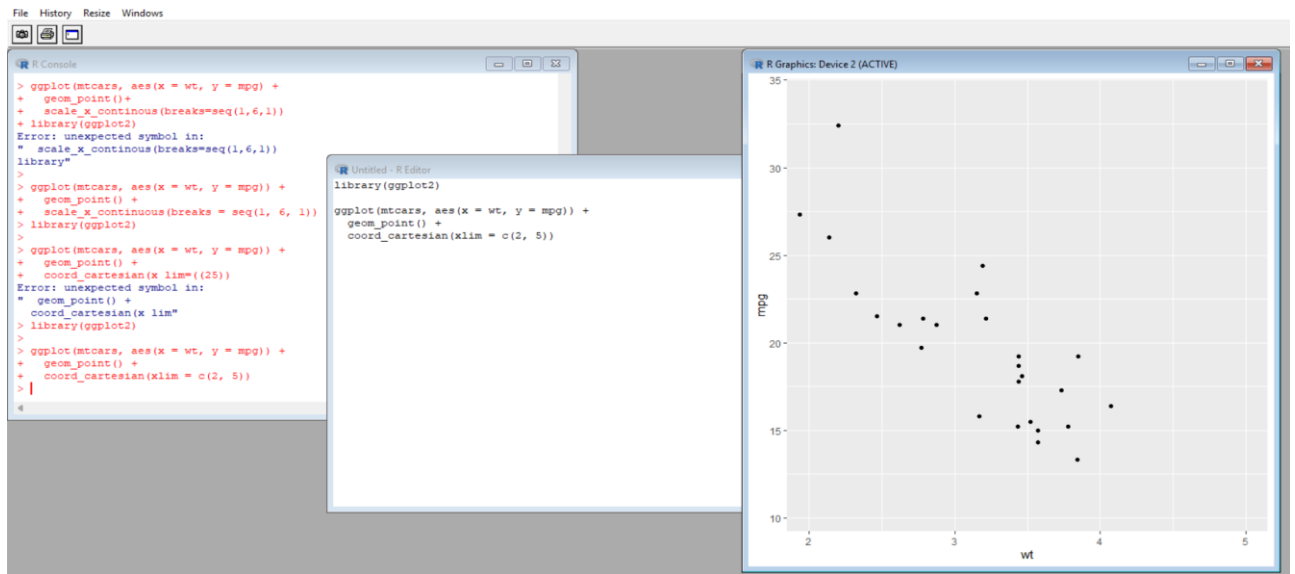
6.cartesian coordinates

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  coord_cartesian(xlim = c(2, 5))
```

Output:



7.log scale

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  scale_y_log10()
```

Output:



8.polar coordinates

Code:

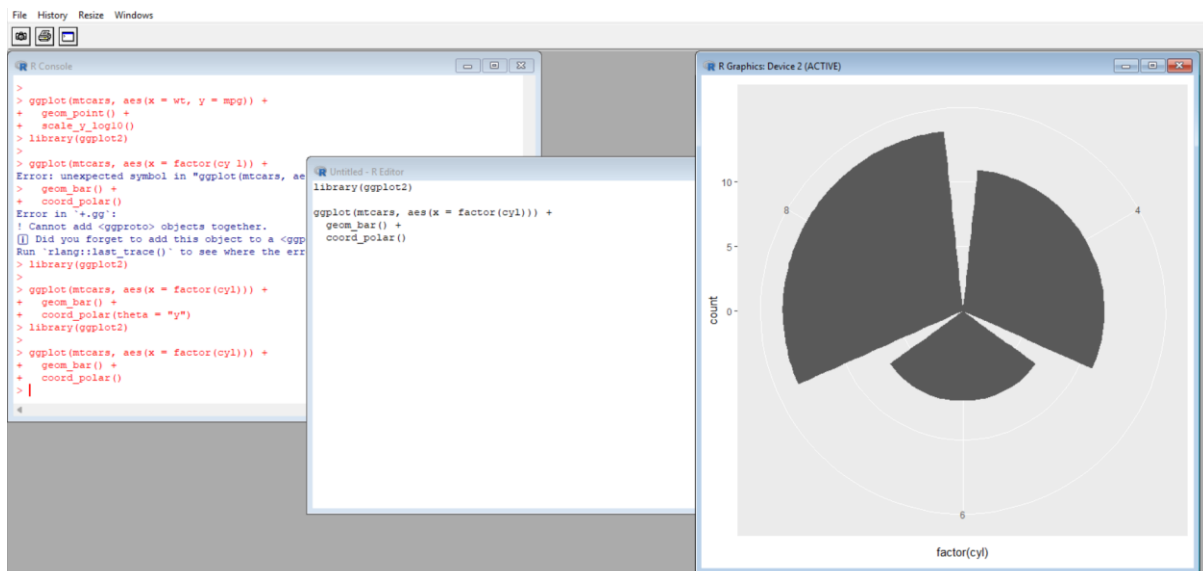
```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = factor(cyl))) +
```

```
  geom_bar() +
```

```
  coord_polar()
```

Output:



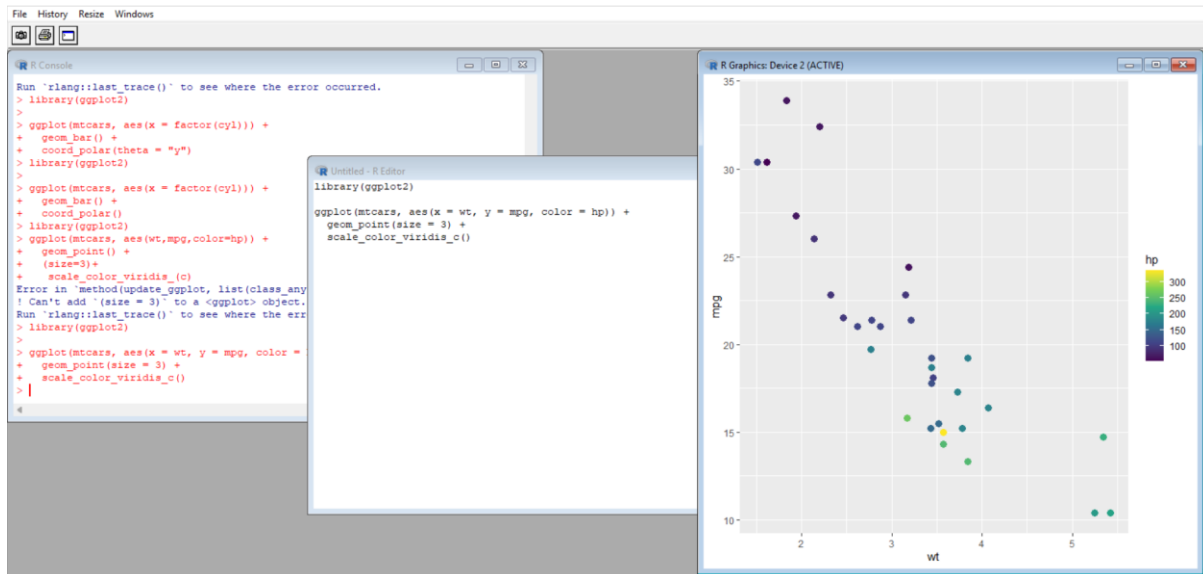
9.sequential color

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, color = hp)) +
  geom_point(size = 3) +
  scale_color_viridis_c()
```

Output:

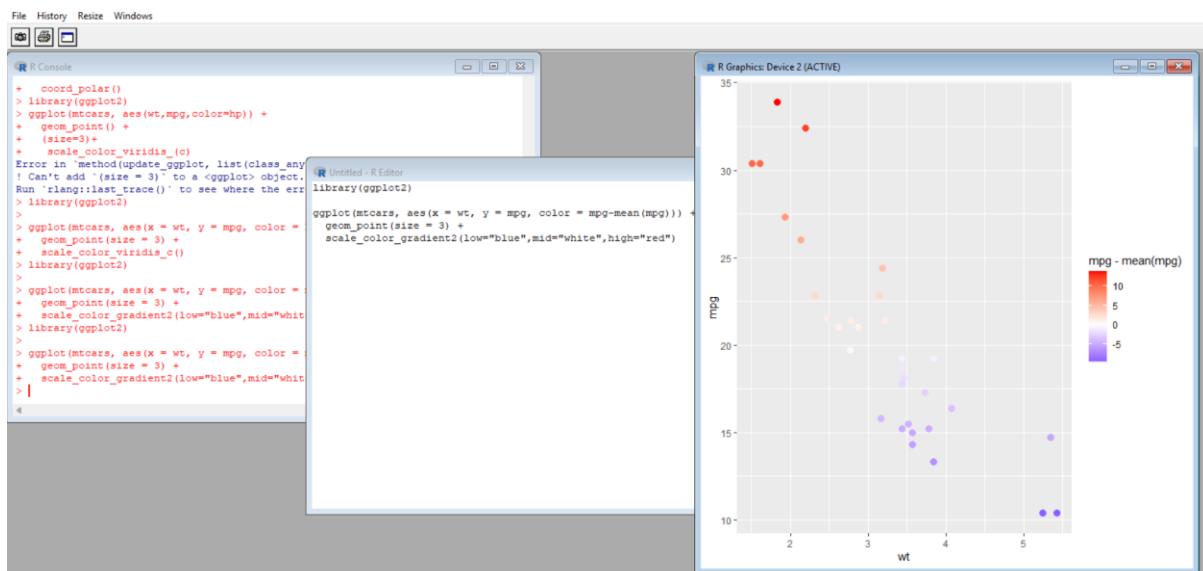


10.Diverging color

```
library(ggplot2)
```

```
ggplot(mtcars, aes(x = wt, y = mpg, color = mpg-mean(mpg))) +
  geom_point(size = 3) +
  scale_color_gradient2(low="blue",mid="white",high="red")
```

Output:



11. color as a tool to distinguish

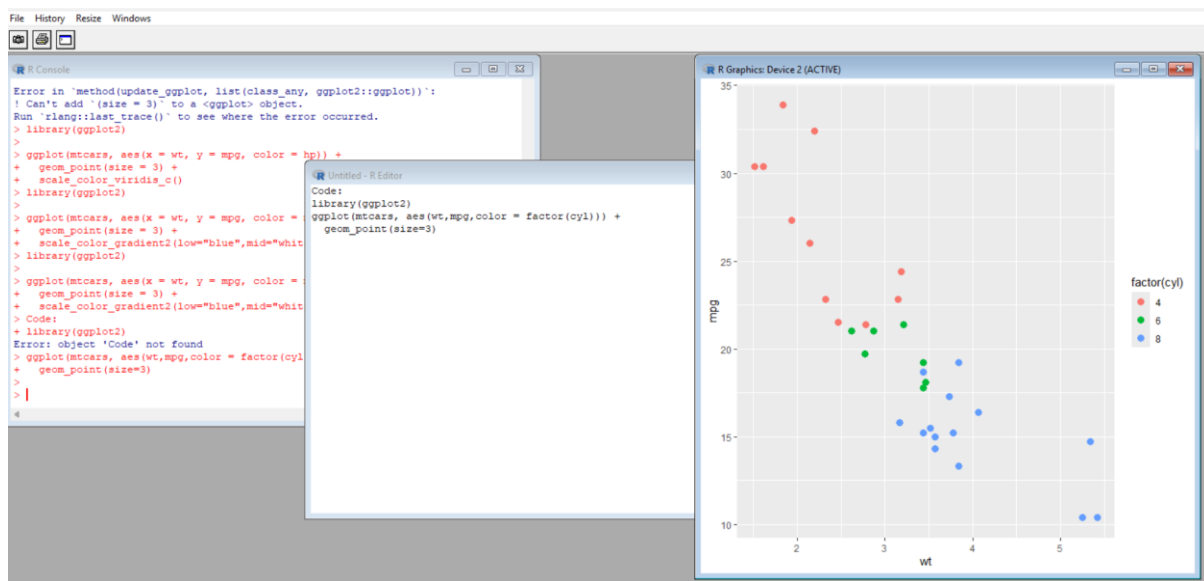
Code:

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(wt, mpg, color = factor(cyl))) +  
  geom_point(size=3)
```

Output:



12. non-linear axis:

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(wt, mpg)) +  
  geom_point() +  
  scale_x_log10() +  
  scale_y_log10()
```

Output:

